DS	**Session**	Searching	**Question Information**	Level 1 Challenge 9

Problem Description:
Prabhu Salamon is planning to make a very long journey across the cityside by Train. His journey consists of N train routes, numbered from 1 to N in the order he must take them. The trains themselves are very fast, but do not run often. The i-th train route only runs every Xi days.

More specifically, he can only take the i-th train on day Xi, 2Xi, 3Xi and so on. Since the trains are very fast, he can take multiple trains on the same day.

Prabhu Salamon must finish his journey by day D, but he would like to start the journey as late as possible. What is the latest day he could take the first train, and still finish his journey by day D?

It is guaranteed that it is possible for Prabhu Salamon to finish his journey by day D.

Constraints:
$1 \leq T \leq 100$.
$1 \leq Xi \leq D$.
$1 \leq N \leq 1000$.
$1 \leq D \leq 10^{12}$

Input Format:
The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with a line containing the two integers N and D. Then, another line follows containing N integers, the i-th one is Xi.

Output Format:
Print the output in a single line contains, the latest day he could take the first train, and still finish his journey by day D.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
3 100
```

Test Case 2

INPUT (STDIN)

```
3
5 1000
```

```
31 71 21
4 100
11 16 51 50
1 1
1
```

EXPECTED OUTPUT

```
62
44
1
```

```
311 171 211 45 32
6 1000
111 161 511 510 56 93
2 10
9 5
```

EXPECTED OUTPUT

```
622
0
9
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(int t=0;t<T;t++)
```

Test Case 2

KEYWORD

```
for(int i=n-1;i>=0;i--)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
4
```

Test Case 2

TOKEN COUNT

```
138
```

Test Case 3

NLOC

```
30
```

Code

solution.c

```c
#include <stdio.h>

int main() {
    int T;
    if (scanf("%d", &T) != 1) return 0;
    for (int t = 0; t < T; t++) {
        int n;
        long long d;
        scanf("%d %lld", &n, &d);
        long long x[n];
        for (int i = 0; i < n; i++) scanf("%lld", &x[i]);
        long long curDay = d;
        for (int i = n - 1; i >= 0; i--) {
            curDay = (curDay / x[i]) * x[i];
        }
        printf("%lld\n", curDay);
    }
    return 0;
}
```

Output

